Simplifying Surds

AQA · KS4 Year 10–11

Name: ____________________ Class: __________ Date: __________

Simplify surds by finding the largest square factor. Express surds in their simplest form a b. Understand why surds are 'exact' values and when to leave answers in surd form.

COMMON MISTAKES

- Not finding the LARGEST square factor — 72 = (36 × 2) = 6 2, not (4 × 18) = 2 18 (not fully simplified).
- Thinking (a + b) = a + b — this is WRONG. (9 + 16) ≠ 9 + 16.
- Writing 4 × 3 = 12 is correct, but the point of simplifying is to go the OTHER way: 12 = 4 × 3 = 2 3.
- Forgetting that a × a = a (not a^2).

METHOD

1. Find the largest perfect square that is a factor of the number under the .
2. Split: n = (square factor × remainder).
3. Use (a × b) = a × b.
4. Take the square root of the perfect square factor out.
5. Write answer as k m where m has no square factors.

WORKED EXAMPLE

Simplify 72.
Step 1 — Largest square factor of 72: 36 (since 36 × 2 = 72).
Step 2 — 72 = (36 × 2)
Step 3 — = 36 × 2
Step 4 — = 6 2
Check: $6^2 \times 2 = 36 \times 2 = 72$

1 Simplify: (2 marks)

a) 12 b) 18 c) 20 d) 50

HOW TO START

Find the largest square number that divides into the number under the root (e.g. 4, 9, 16, 25, 36...).

Split it: 12 = (4 × 3) = 4 × 3.

Take the square root of the square factor: 4 = 2, so 12 = 2 3.

Answer ____________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

2 Simplify: (2 marks)

a) 8 b) 32 c) 45 d) 75

HOW TO START

Same method: largest square factor first.

For 8, the largest square factor is 4: 8 = 4 × 2 = 2 2.

Answer ______________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

3 Simplify: (3 marks)

a) 98 b) 200 c) 128 d) 300

Answer ____________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

4 Simplify: (3 marks)

a) (4 × 7) b) 3 12 c) 2 50 d) 5 8

Answer ____________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

5 Simplify fully: (3 marks)

a) 48 + 12 b) 75 - 27 c) 2 18 + 3 2

Answer ______________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

6 Show that: (4 marks)

a) 80 = 4 5 b) 3 28 = 6 7 c) 180 = 6 5

Answer ____________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

7 A square has area 72 cm^2. (4 marks)

a) Find the exact side length. (1 mark)
b) Find the exact perimeter. (1 mark)
c) A second square has side length 3 2 cm. Which square is larger? Show working. (2 marks)

Answer ______________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

8 a) Show that $(a^2b) = a \quad b$ for any positive a and b. (1 mark) (5 marks)

b) Simplify $(50x^4y^2)$ where $x, y > 0$. (2 marks)

c) A right-angled triangle has shorter sides 18 cm and 32 cm. Find the exact hypotenuse in simplest surd form. (2 marks)

Answer ____________________

METHOD 1 Find largest square factor | 2 Split under the root: (square × rest) | 3 Use (a × b) = a × b | 4 Square-root the square factor | 5 Answer: k m (m has no square factors)

Self-reflection

How confident do you feel with this topic now? (tick one)

Not yet	Getting there	Confident	Really confident

Which question did you find the hardest, and why?

Do I always find the LARGEST square factor first time?

One thing I will practise before the next lesson:

Teacher key

Q1: a) 2 3 b) 3 2 c) 2 5 d) 5 2

Q2: a) 2 2 b) 4 2 c) 3 5 d) 5 3

Q3: a) 7 2 b) 10 2 c) 8 2 d) 10 3

Q4: a) 2 7 b) 3 × 2 3 = 6 3 c) 2 × 5 2 = 10 2 d) 5 × 2 2 = 10 2

Q5: a) 4 3 + 2 3 = 6 3 b) 5 3 - 3 3 = 2 3 c) 6 2 + 3 2 = 9 2

Q6: a)   80 =   (16×5) = 4 5   b) 3 28 = 3 (4×7) = 3×2 7 = 6 7   c)   180 =   (36×5) = 6 5

Q7: a)   72 = 6 2 cm b) 4 × 6 2 = 24 2 cm c) Second square side = 3 2, first = 6 2. 6 2 > 3 2, so first is larger.

Q8: a)   (a^2b) =   (a^2) ×   b = a b   b)   (50x^4y^2) =   (25 × 2 × x^4 × y^2) = 5x^2y 2 c) Hyp^2 = 18 + 32 = 50; hyp =   50 = 5 2 cm